

QUICK-COMMERCE MADE SIMPLE

Stack: React Native • FastAPI • Supabase

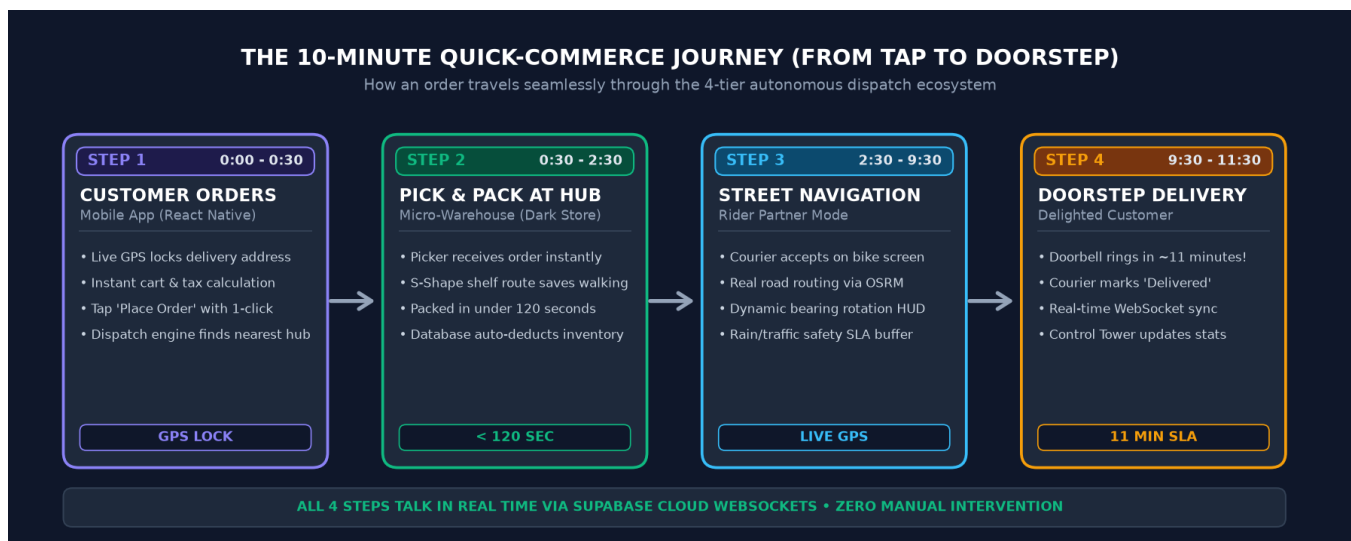
Live Repo: github.com/NeelBelsare

Delivery SLA: Sub-12 Min Guaranteed

Visual Non-Tech Handbook & How-To Guide (v3.0.0)

Welcome! You don't need any programming or mathematical background to understand this project. This visual manual explains how 10-minute grocery delivery apps (like Blinkit, Zepto, and Instamart) actually operate, how our system was engineered to deliver across Chhatrapati Sambhajnagar (Aurangabad), and provides an easy 4-step walkthrough with clickable links and screenshots so you can test everything yourself.

Visual Flowchart: The 10-Minute Order Lifecycle



Flowchart 1: The 4-step journey of an order — from the instant you tap 'Order' on your phone to delivery at your doorstep.

What is a 'Dark Store'? (It's not spooky!): Traditional supermarkets take days because giant warehouses are 50 km outside the city. Quick-Commerce places compact mini-warehouses directly inside residential neighborhoods. They have **no walk-in shoppers, glass display windows, or long cashier queues**. Because only trained pickers are inside, your grocery bag is packed in **under 120 seconds** and handed to a waiting bike courier!

1. The 4 System Pillars & Cloud Architecture

Our platform connects four powerful tiers in real time. Click any link below to inspect the actual code on GitHub:

Tier	Everyday Analogy	What It Does & GitHub Link
1. Mobile App React Native Expo	Storefront & Bike Screen	Browse groceries, see live bills, and place orders. Switch to 'Rider Mode' for street navigation. ■ View mobile-app/ code
2. Dispatch Engine FastAPI Backend	Traffic Police & Matchmaker	Finds the closest dark store, assigns couriers, and calculates weather/traffic buffers. ■ View api.py backend
3. Cloud Memory Supabase PostgreSQL	Master Ledger & Counter	Tracks real-time stock levels, store locations, and active orders via WebSockets. ■ View Database Schema
4. Control Tower Streamlit Dashboard	Airport Control Tower	Visual screen for managers: 3D flight arcs, moving couriers, and 1-click restock. ■ View app.py dashboard

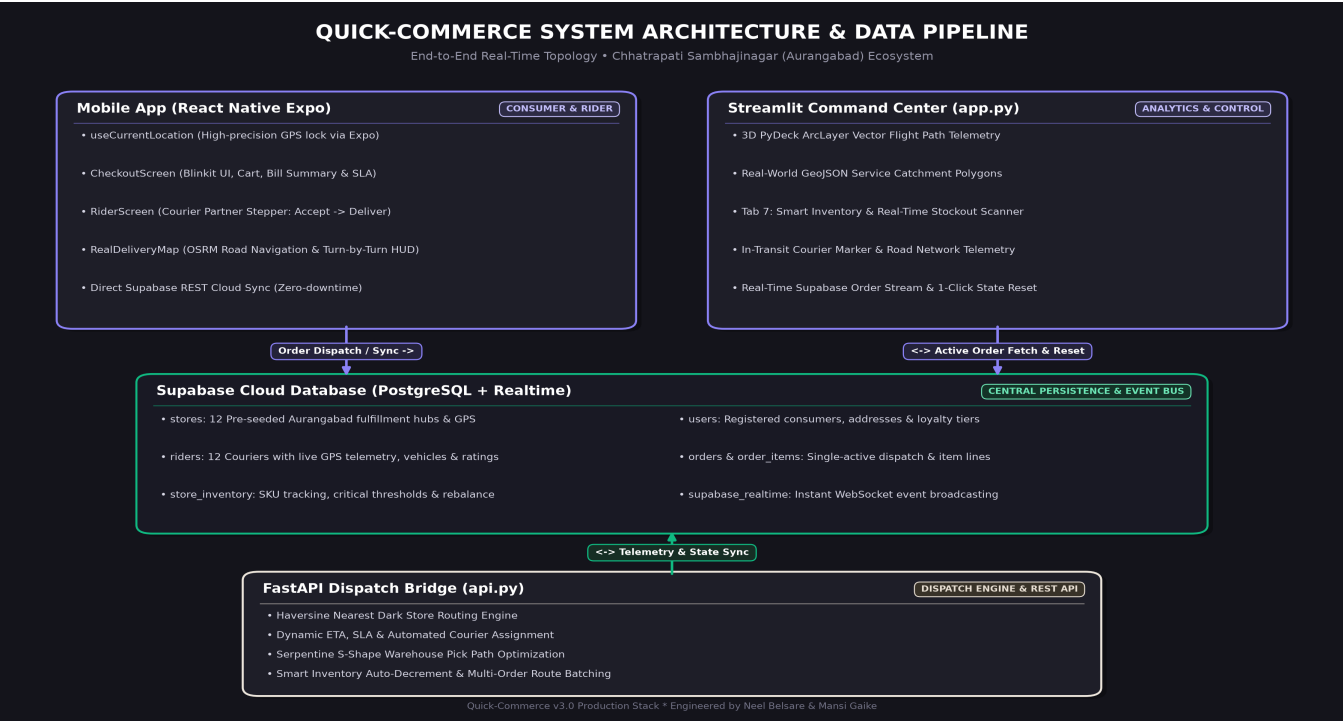


Figure 1: System Architecture Diagram — Showing how customer phones, cloud databases, and manager dashboards connect seamlessly.

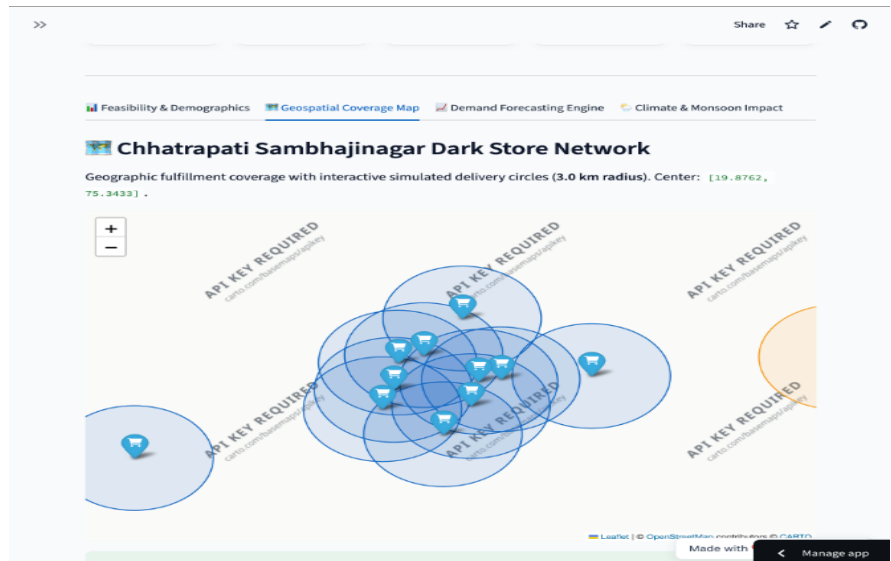
2. Visual Walkthrough & Real Application Screenshots

Here are actual screenshots taken directly from the running applications:



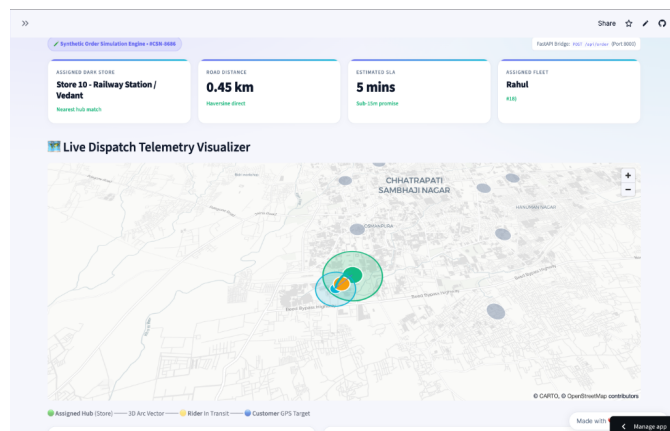
Mobile App: Checkout & Rider Partner Road HUD.

■ [Explore Mobile App Code](#)



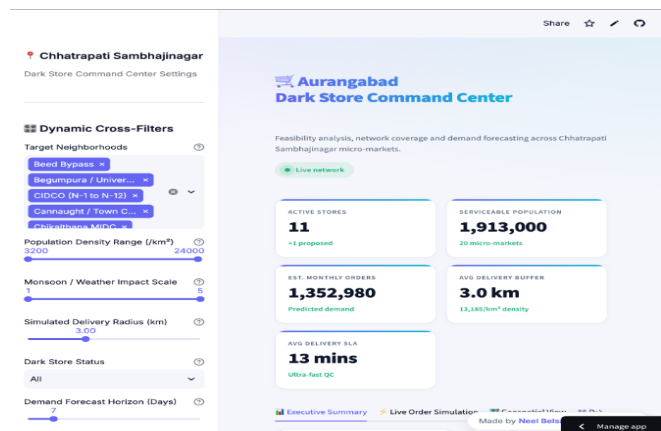
12 Dark Stores: City Coverage Zones in Aurangabad.

■ [See 12 Hub Locations](#)



3D Flight Telemetry: Live in-transit rider arcs.

■ [PyDeck 3D Features](#)



KPIs & Economics: Profits, SLAs, and order volume.

■ [Analytics Engine](#)

3. Step-by-Step Guide: How to Test the Project

You can run and test every component locally or through Docker in 4 easy steps:

Quick Start (For Anyone with Terminal or Docker):

- **Option A (Easiest - 1 Command via Docker):** Run docker compose up in your terminal.
 - **Option B (Standard):** Run streamlit run app.py to launch the Manager Control Tower.
 - **Option C (Mobile App):** Open mobile-app/ and run npx expo start to test on iOS, Android, or web browser.
- **Full Setup Guide:** <https://github.com/NeelBelsare/my-dark-store-app#quick-start>

Step 1: Open the Manager's Command Center (Browser)

Open your web browser (Chrome/Safari) to **http://localhost:8501**. You will see the sleek Lavender & Dark Slate dashboard.

- Click **'Live 3D Map'** to watch real-time flight arcs and courier markers.
- Click **'Smart Inventory'** (Tab 7) to view live warehouse stock levels.

Step 2: Place an Order as a Customer (Mobile Screen)

Open the mobile app or simulator screen.

- Tap **'+'** on snacks or groceries (e.g. Potato Chips, Fresh Milk, Soda).
- Review the instant bill summary with tax calculation and celebratory free-delivery progress bar.
- Tap **'Place Order'** — within 0.1 seconds, celebratory confetti bursts and the order is locked!

Step 3: Switch to 'Rider Partner Mode' (Experience the Delivery)

Tap the **'Rider Mode'** button at the top of the mobile screen to see through the courier's eyes:

- Follow the 4-stage stepper: *Accept Order* → *Pick & Pack at Store* → *Out for Delivery* → *Delivered*.
- Watch the live blue route map: it follows actual streets (Jalna Road, Kranti Chowk, CIDCO) and turns the bike icon as you move!

Step 4: Watch the Inventory Automatically Drop

Switch back to the Manager's Dashboard under **Tab 7: Smart Inventory**:

- The item you ordered will decrease by exactly 1 unit in real time via cloud sync.
- If an item drops below 10 units, a bright red **'LOW STOCK ALERT'** badge appears.
- Tap the **'Auto-Replenish'** button to instantly balance inventory from a neighboring hub!

4. Frequently Asked Questions (FAQ for Non-Techies)

Q1: Why 12 mini-stores instead of 1 giant supermarket?

Driving across Aurangabad takes 40+ minutes in heavy traffic. By opening 12 compact mini-stores across Cidco, Kranti Chowk, Jalna Road, and Waluj, no customer is ever more than 2.5 km away from a store!

Q2: What happens if it rains heavily in Aurangabad?

Delivery couriers should never have to drive dangerously fast. Our system automatically checks live weather and adds a 3-to-5 minute safety buffer to the estimated arrival time, keeping riders safe while keeping customers informed.

Q3: How do warehouse workers find items so quickly?

Inside regular stores, shoppers wander around aisles. In our dark stores, an intelligent 'S-Shape Picking Route' tells the worker the exact sequence of shelves to walk through, like walking in an 'S' curve, so they never backtrack.

Q4: Can this be used for other cities or products?

Yes! The mapping engine is completely modular. You can enter coordinates for Pune, Nashik, or Mumbai, or use it for rapid medicines, fresh meats, or pet supplies in minutes.

5. Interactive Links & Project Credits

Core Author & Systems Lead:	Neel Belsare • Full-Stack & Systems Engineer ■ Email: neelbelsare28@gmail.com ■ LinkedIn: linkedin.com/in/neel-belsare ■ GitHub: github.com/NeelBelsare
Co-Author & Analytics Lead:	Mansi Gaike • Data Science & Feasibility Analyst
Live Open-Source Repository:	■ https://github.com/NeelBelsare/my-dark-store-app Tagged: Version 3.0.0 Production Release